

Table of contents

Relative Entropy	1
Question	1
Introduction to Relative Entropy (Kullback-Leibler Divergence)	2
Definition and Intuition	2
Fundamental Properties	2
1. Non-negativity	2
2. Asymmetry	2
3. Convexity	2
4. Additivity for Independent Distributions	2
Relations with Other Information Measures	3
Link with Cross-Entropy	3
Link with Mutual Information	4
The Information Triangle: Complete Relations	6
Examples of Practical Use	6
1. Machine Learning: Loss Function	6
2. Data Compression: Suboptimal Codes	7
3. Variational Inference	7
Summary Table of Relations	7
Complete Pedagogical Example: The Detective and Their Hypothesis	8
Advanced Applications	8
1. Regularization via KL Divergence	8
2. Reinforcement Learning	9
3. Bayesian Information Theory	9
Critical Points of Attention	10
1. Asymmetry and Direction Choice	10
2. Sensitivity to Rare Events	10
3. Symmetric Alternatives	11
Synthetic Conclusion	11

Relative Entropy

Question

Explain relative entropy, its properties, and give some examples of its use.

Introduction to Relative Entropy (Kullback-Leibler Divergence)

Relative entropy, also called **Kullback-Leibler divergence (KLD)**, is a fundamental measure in information theory that quantifies the difference between two probability distributions. It plays a central role in relation to mutual information and cross-entropy.

Definition and Intuition

For two discrete distributions $p(x)$ and $q(x)$ defined on the same space $\mathcal{X}$:

$$D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log_2 \frac{p(x)}{q(x)} = \mathbb{E}_{p(x)} \left[\log_2 \frac{p(x)}{q(x)} \right]$$

Intuitive interpretation: KLD measures the **average number of extra bits** required to encode samples coming from distribution p when using a code optimized for distribution q .

Fundamental Properties

1. Non-negativity

$$D_{\text{KL}}(p \parallel q) \geq 0$$

Equality $D_{\text{KL}}(p \parallel q) = 0$ occurs if and only if $p = q$ almost everywhere.

2. Asymmetry

$$D_{\text{KL}}(p \parallel q) \neq D_{\text{KL}}(q \parallel p)$$

KLD is **not a metric distance** because it is not symmetric and does not satisfy the triangle inequality.

3. Convexity

KLD is convex with respect to each argument, which is crucial for optimization problems.

4. Additivity for Independent Distributions

If $p(x, y) = p_1(x)p_2(y)$ and $q(x, y) = q_1(x)q_2(y)$, then:

$$D_{\text{KL}}(p \parallel q) = D_{\text{KL}}(p_1 \parallel q_1) + D_{\text{KL}}(p_2 \parallel q_2)$$

Relations with Other Information Measures

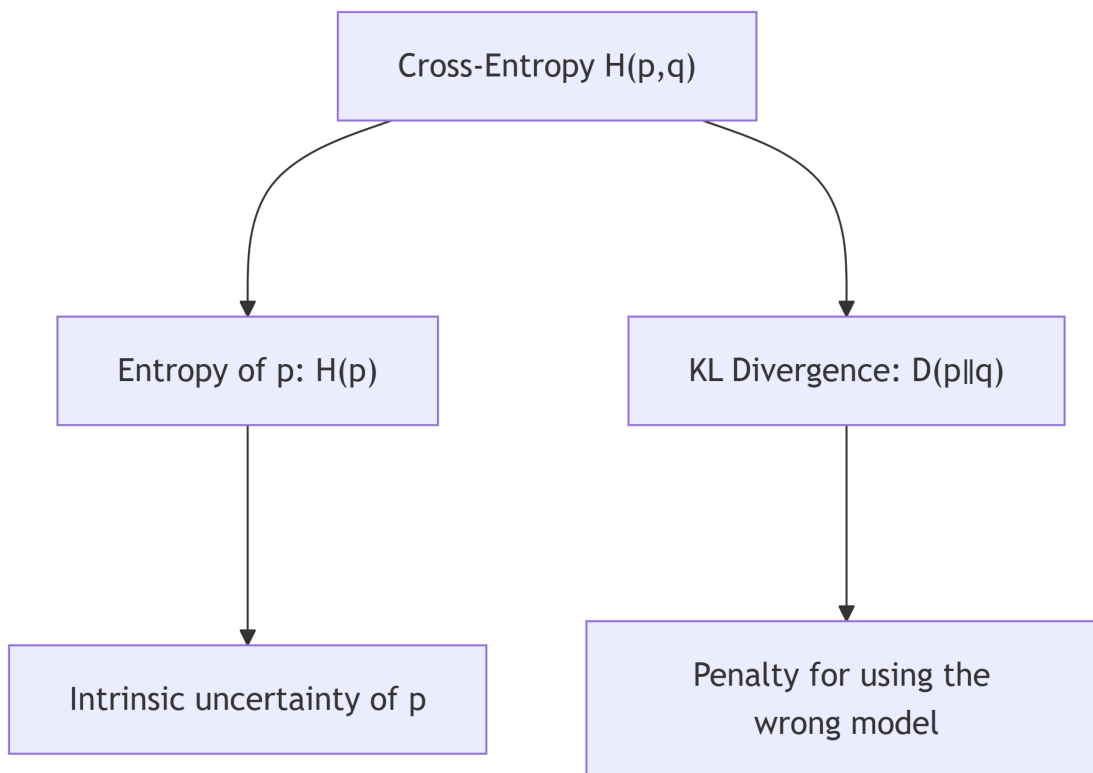
Link with Cross-Entropy

Definition of cross-entropy:

$$H(p, q) = - \sum_x p(x) \log_2 q(x) = \mathbb{E}_{p(x)}[-\log_2 q(x)]$$

Fundamental relation:

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q)$$



Pedagogical interpretation:

- $H(p)$: Optimal number of bits to encode p (with a perfectly adapted code)
- $D_{\text{KL}}(p \parallel q)$: Overhead due to using the code for q to encode p
- $H(p, q)$: Total bits used = optimal + overhead

Concrete example:

Suppose a student must study for an exam. Let:

- p : Real distribution of questions (unknown to the student)
- q : Distribution assumed by the student (based on their studying)
- $H(p)$: Real difficulty of the exam
- $D_{\text{KL}}(p \parallel q)$: The student's "surprise" when facing the actual questions
- $H(p, q)$: Difficulty perceived by the student

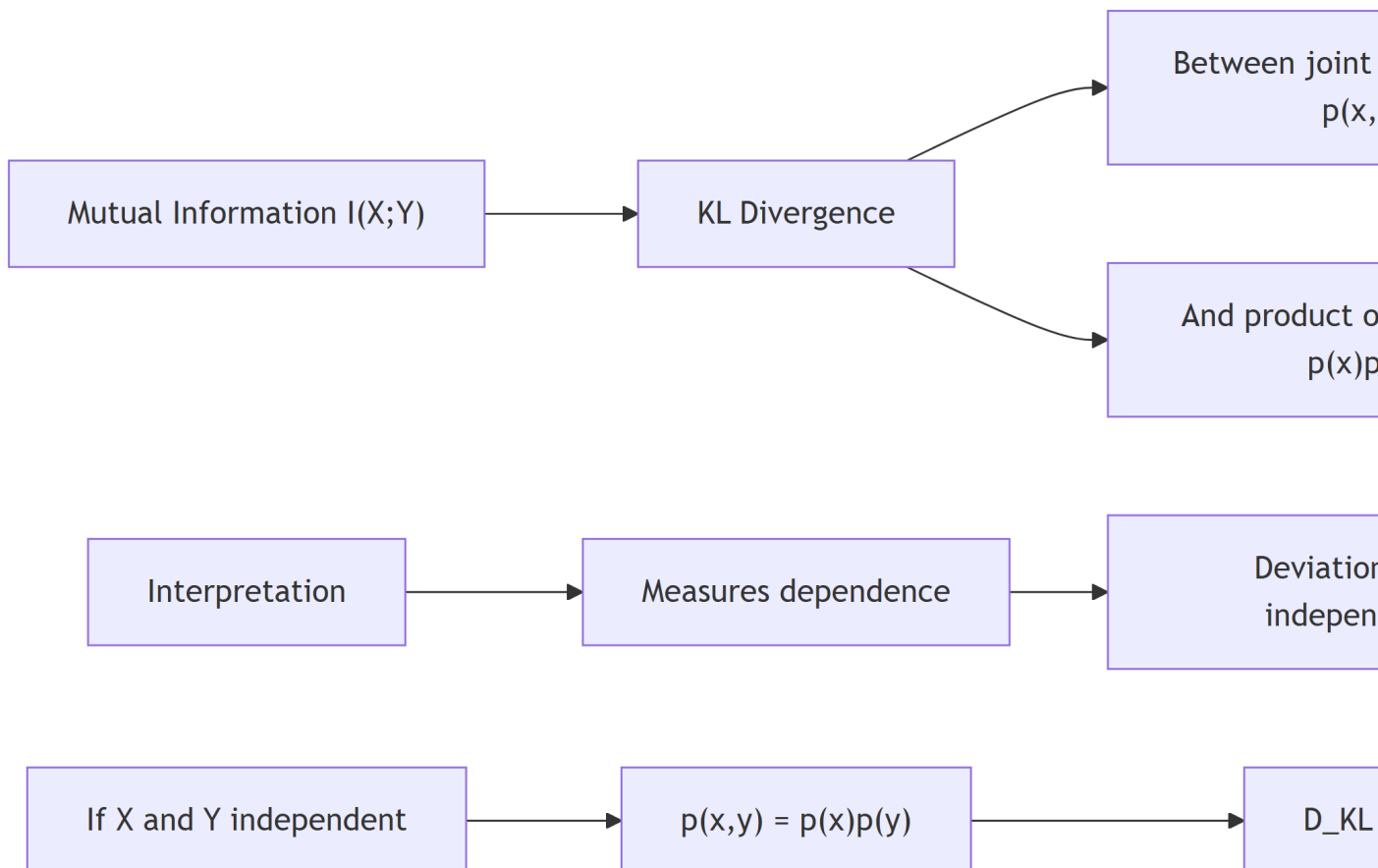
Link with Mutual Information

Recall of mutual information:

$$I(X; Y) = \sum_{x, y} p(x, y) \log_2 \frac{p(x, y)}{p(x)p(y)}$$

Fundamental relation:

$$I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y))$$

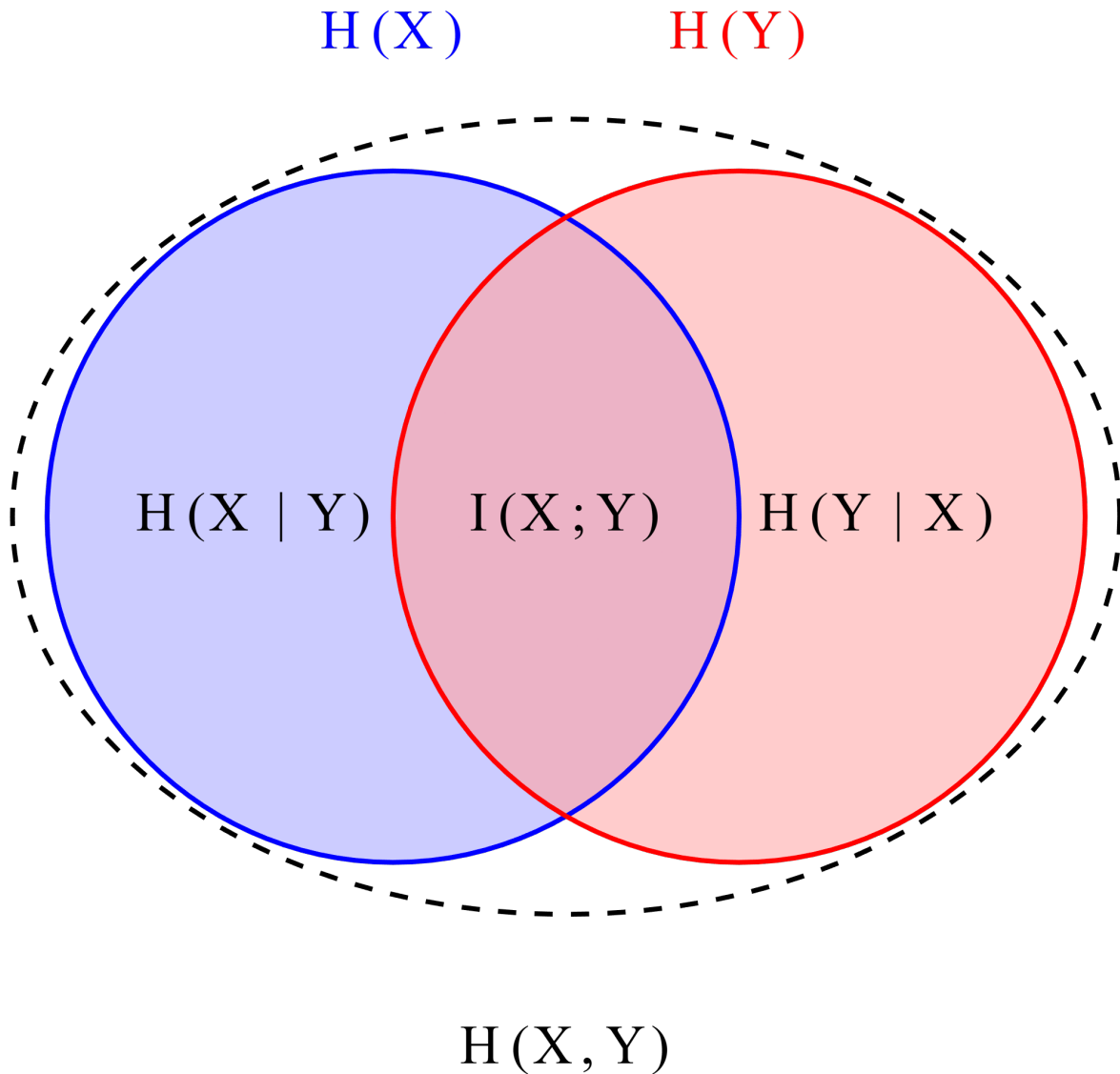


Pedagogical interpretation:

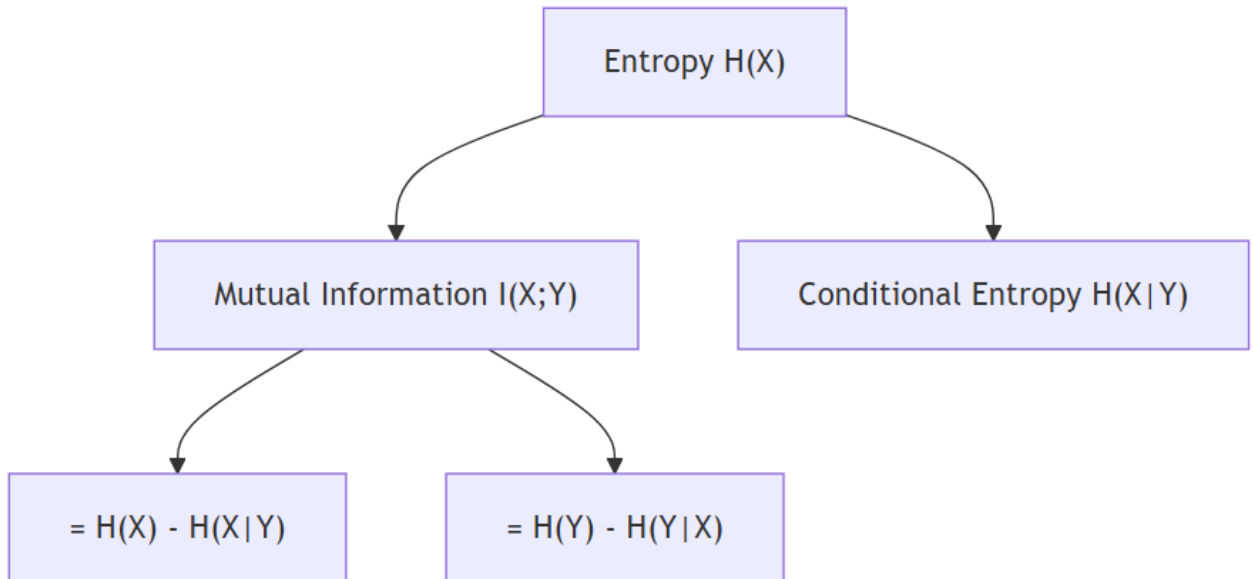
Mutual information measures “how much we learn about X by observing Y ” (and vice-versa). It is the KL divergence between:

- Reality: joint distribution $p(x, y)$ (with potential dependence)
- Independence hypothesis: $p(x)p(y)$ (if we think X and Y are independent)

Visual example:



The Information Triangle: Complete Relations



Examples of Practical Use

1. Machine Learning: Loss Function

In classification, **cross-entropy loss** is often used:

Context:

- p : Target distribution (one-hot encoding of labels)
- q : Distribution predicted by the model (softmax)
- We minimize $H(p, q)$ to train the model

Why it works: Minimizing $H(p, q)$ is equivalent to minimizing $D_{\text{KL}}(p \parallel q)$ because $H(p)$ is constant (the labels are fixed).

Numerical example: For binary classification with true label = 1:

- $p = [0, 1]$ (target distribution)
- If the model predicts $q = [0.1, 0.9]$: $H(p, q) = -\log_2(0.9) \approx 0.152$ bits
- If the model predicts $q = [0.4, 0.6]$: $H(p, q) = -\log_2(0.6) \approx 0.737$ bits

2. Data Compression: Suboptimal Codes

Scenario: We design a code for a language with distribution q , but we use it for a language with distribution p .

Example:

- Language A (p): ‘e’=12.7%, ‘t’=9.1%, ‘z’=0.07%
- Language B (q): ‘e’=10.0%, ‘t’=15.0%, ‘z’=1.0%
- Optimal code for B assigns short codes to ‘t’ and ‘z’

Calculation:

$$D_{\text{KL}}(p \parallel q) = 0.127 \log_2 \frac{0.127}{0.10} + 0.091 \log_2 \frac{0.091}{0.15} + \dots$$

This divergence measures the inefficiency of code B for language A.

3. Variational Inference

Problem: Approximate a complex posterior distribution $p(z|x)$ with a simple distribution $q(z)$.

Approach: Minimize $D_{\text{KL}}(q \parallel p)$ so that q is close to p .

Direction choice:

- $D_{\text{KL}}(q \parallel p)$: “mode-seeking” (avoids areas where $q > 0$ but $p = 0$)
- $D_{\text{KL}}(p \parallel q)$: “mean-seeking” (covers all modes of p)

Summary Table of Relations

Concept	Formula	Relation with KLD	Interpretation
Cross-Entropy	$H(p, q) = -\sum p \log q$	$H(p, q) = H(p) + D(p \parallel q)$	Cost of using the wrong model
Mutual Information	$I(X; Y) = \sum p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$	$I(X; Y) = D(p(x, y) \parallel p(x)p(y))$	Dependence between variables
Conditional Entropy	$H(X \parallel Y) = H(X, Y) - H(Y)$	$H(X \parallel Y) = H(X) - I(X; Y)$	Residual uncertainty
Joint Entropy	$H(X, Y) = -\sum p(x, y) \log p(x, y)$	$H(X, Y) = H(X) + H(Y) - I(X; Y)$	Total uncertainty

Complete Pedagogical Example: The Detective and Their Hypothesis

Scenario: A detective investigates a series of crimes.

- $p(x)$: Real distribution of the true criminal's characteristics
 - Age: 30-40 years (40%), 40-50 years (40%), other (20%)
 - Vehicle: Car (70%), Motorcycle (30%)
- $q(x)$: Detective's hypothesis
 - Age: 20-30 years (60%), 30-40 years (30%), other (10%)
 - Vehicle: Car (50%), Motorcycle (50%)

KL Divergence calculation:

For each characteristic, we calculate:

$$D_{\text{KL}}(p \parallel q) = \sum p(x) \log_2 \frac{p(x)}{q(x)}$$

Interpretation:

- If D_{KL} is large $\rightarrow$ the detective has a poor hypothesis
- If $D_{\text{KL}} = 0 \rightarrow$ the detective guessed the distribution perfectly

Link with mutual information:

Suppose we now discover a relationship between:

- X : Type of crime
- Y : Vehicle used

The mutual information $I(X; Y)$ measures whether knowing the type of crime gives information about the vehicle (and vice-versa).

Advanced Applications

1. Regularization via KL Divergence

In generative models (like VAEs), we use $D_{\text{KL}}(q(z|x) \parallel p(z))$ to constrain the latent distribution to follow a prior distribution $p(z)$ (often Gaussian).

2. Reinforcement Learning

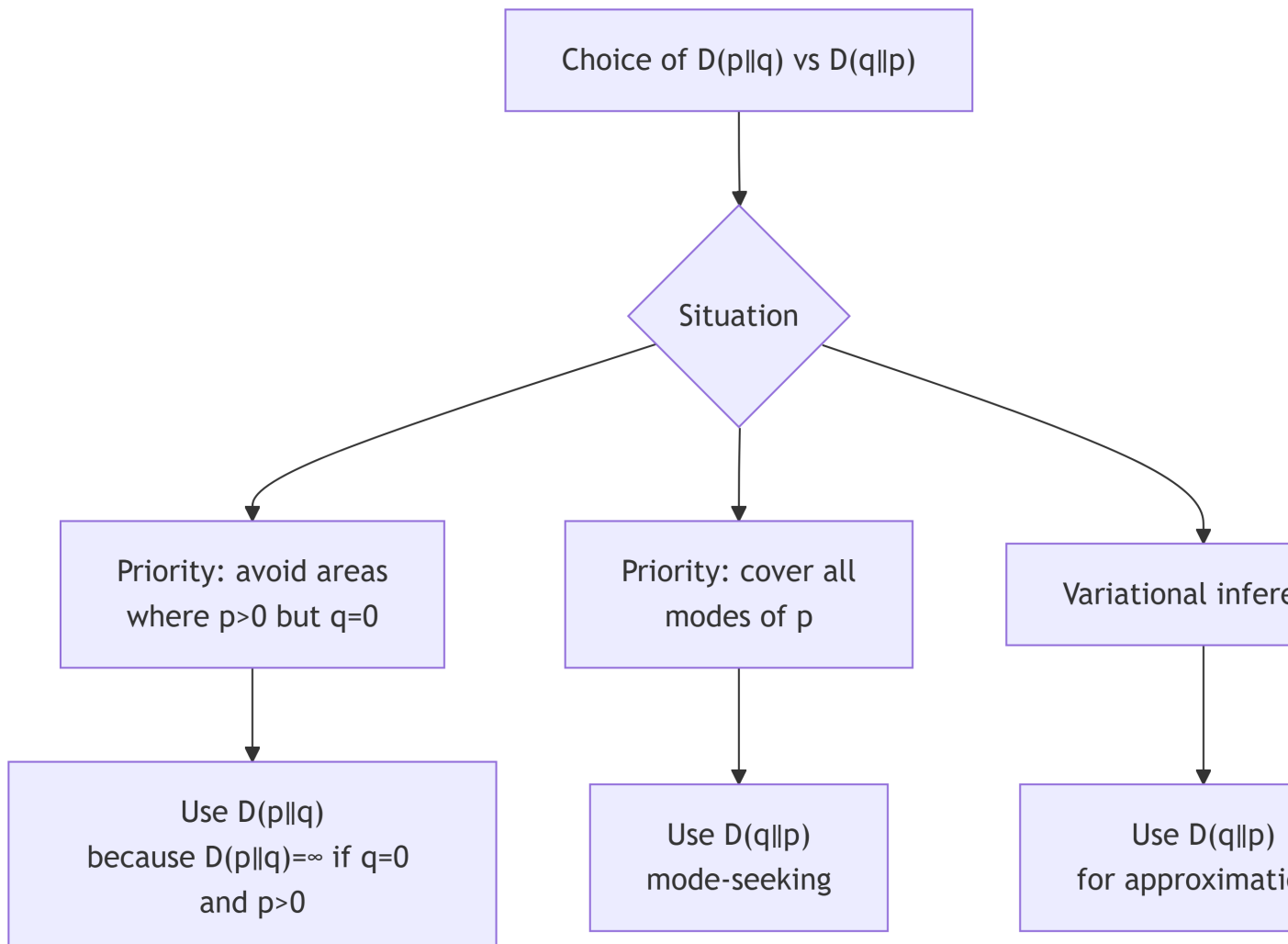
KL divergence is used in algorithms like TRPO and PPO to constrain policy updates, preventing overly drastic changes.

3. Bayesian Information Theory

In Bayesian inference, $D_{\text{KL}}(p_{\text{post}} \parallel p_{\text{prior}})$ measures how much observing the data has modified our beliefs.

Critical Points of Attention

1. Asymmetry and Direction Choice



2. Sensitivity to Rare Events

KLD is very sensitive to low-probability events in p that are zero in q :

- $p(x) > 0$ but $q(x) = 0 \rightarrow D_{\text{KL}}(p \parallel q) = \infty$

3. Symmetric Alternatives

When symmetry is needed:

- **Jensen-Shannon divergence:** $JS(p, q) = \frac{1}{2}D(p \parallel m) + \frac{1}{2}D(q \parallel m)$ where $m = \frac{p+q}{2}$
- **Wasserstein distance:** Based on optimal transport

Synthetic Conclusion

Relative entropy (KLD) is the **conceptual cement** that links the main information measures:

1. **With cross-entropy:** $H(p, q) = H(p) + D(p \parallel q)$
 - KLD is the “overhead” when using a wrong model
2. **With mutual information:** $I(X; Y) = D(p(x, y) \parallel p(x)p(y))$
 - KLD measures the deviation from independence
3. **With conditional entropy:** $H(X|Y) = H(X) - I(X; Y)$
 - The reduction in uncertainty thanks to Y

Final metaphor: Imagine you have to guess the contents of a box:

- $H(p)$: Number of questions needed if you know the true distribution
- $H(p, q)$: Number of questions if you have an imperfect hypothesis q
- $D(p \parallel q)$: Extra questions due to your poor hypothesis
- $I(X; Y)$: Questions saved by listening to a clue Y

Thus, KL divergence quantifies the “cost of ignorance” - how much we pay in efficiency when our models do not match reality.